

MOP: A Multimodal Object-aware Policy for Robotic Manipulation via Geometry-guided Fusion and Trajectory Prediction

Abstract—3D imitation learning has demonstrated capability in diverse visuomotor tasks but often struggles with small objects or high-precision manipulation due to geometric sparsity. To address this, we propose the Multimodal Object-aware Policy (MOP), a novel framework that incorporates a geometry-guided fusion module to adaptively integrate 2D semantic features with 3D geometry for precise control. Additionally, we introduce a lightweight Object Position Prediction (OPP) module that serves as an auxiliary supervision signal. The training labels for this module are generated using a cost-effective vision-based software tracker, effectively replacing expensive hardware-based motion capture systems. We evaluate our policy across 56 tasks on 3 simulation benchmarks. Experimental results demonstrate that MOP significantly outperforms baselines, achieving higher success rates with lower variance and efficient inference. Real-world experiments on 4 tasks further validate the robustness and transferability of our approach. The project resources are available at <https://reshmouqi01.github.io/MOPpage/>.

Index Terms—Learning from Demonstration, Deep Learning in Grasping and Manipulation, Perception for Grasping and Manipulation.

I. INTRODUCTION

IN the dynamic and unstructured real-world environments, imitation learning enables the robots to learn human-like manipulation skills. To better understand the physical world, robotic perception systems play an essential role in the embodied intelligence. By incorporating 2D visual perception, transformer-based and diffusion-based imitation learning architectures are adopted to predict the desired actions [1, 2]. Recent works have shown that leveraging 3D representations with point clouds [3–5] and voxels [6] can significantly improve the robotic manipulations in terms of accuracy and robustness, compared to the 2D counterpart.

Prior 3D diffusion strategies are capable of solving conventional manipulation tasks, however, when confronted with small objects¹ or high-precision tasks [7], which include not only fine-grained assembly but also contact-sensitive dynamic interactions (e.g., push-grasp), they may not efficiently address these challenging tasks due to limited spatial understanding. In such tasks, inaccurate geometric perception can lead to task failure. As such, rich 3D semantics and accurate geometry are required in more complex environments. Recent work [8] adopts NeRF for pre-training to learn 3D representations, in order to accomplish difficult robotic manipulation. To redeem the insufficient geometric information, the pre-trained large vision-language models [9, 10] or language-guided high-level planning [11, 12] are leveraged to understand the semantics, but these approaches could not guarantee low-level geometric alignment. Existing contributions such as [13, 14] fuse the 2D

¹In this work, an object is regarded as ‘small’ not strictly by an absolute physical bounding box, but primarily by its perceptual sparsity, i.e., its point cloud is highly sparse.

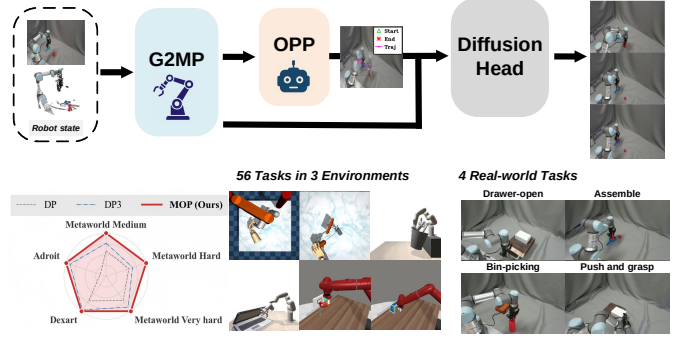


Fig. 1: Multimodal object-aware policy (MOP) is a multimodal imitation learning method, enabling high accuracy and strong robustness on various simulation benchmarks and real-world tasks.

semantics with 3D geometry for spatial feature alignment, and consider point cloud and image features to be equally important. Such consideration neglects dynamic modality combinations in favor of dense semantics, resulting in reduced accuracy and computational efficiency. Therefore, adaptive multimodal fusion is an important means to enhance efficiency and robustness [15]. In robotic manipulation tasks, leveraging object motion priors is crucial for integrating perception with control. To capture fine-grained motion cues, some approaches such as transferable affordance [16], trajectory modeling for videos [17] and 3D scene flow [18] have been studied in the literature. With the help of an expensive motion capture (MoCap) system, [19] designs two cascaded diffusion processes for object pose prediction and action generation, which limits the system scalability and inference speed.

To address the aforementioned issues, this work proposes a multimodal object-aware policy (MOP) (Fig. 1), which is a novel imitation learning framework. Different from existing designs, MOP develops a geometry-guided multimodal perception module and a lightweight object position prediction module that replaces the reliance on expensive hardware-based MoCap systems with low-cost vision-based tracking algorithm (e.g., Cotracker [20]) during training data preparation. To comprehensively assess MOP performance, we conduct simulations across 56 tasks on 3 simulation benchmarks including Adroit [21], DexArt [7] and Meta-World [22]. The results confirm that MOP substantially improves the performance and accelerates inference, compared to the baseline methods. The real-robot experiments further demonstrate the efficacy of our method. The main contributions of this work are as follows:

- We propose a geometry-guided multimodal perception (G2MP) module to enable robust multimodal perception. Through employing an asymmetric architecture with adaptive gating, G2MP dynamically regulates the injection of RGB semantics into 3D geometric representations. This mechanism upweights visual representations to com-

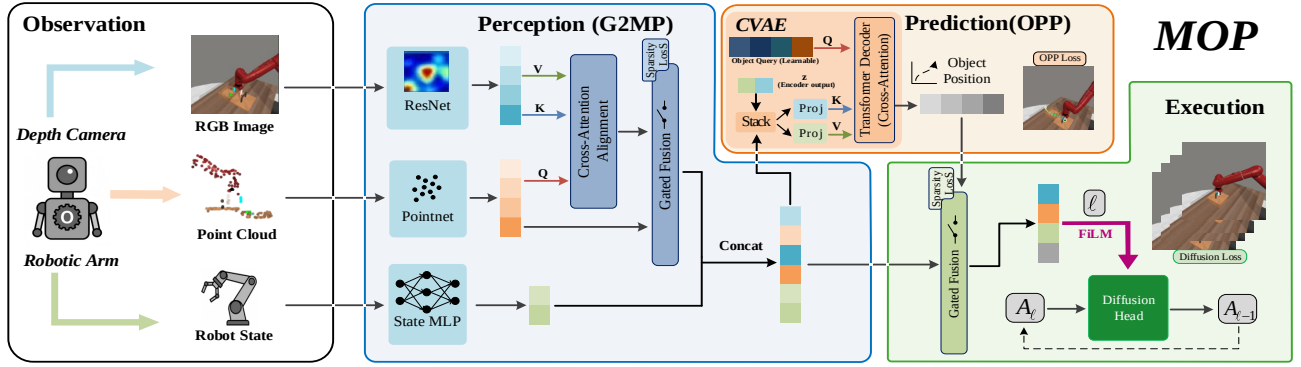


Fig. 2: **Overview of the multimodal object-aware policy (MOP).** Based on observations captured by RGB-D camera and robot state, MOP adopts the geometry-guided multimodal perception (G2MP) for dynamically fusing the semantics and geometric information. In the object position prediction (OPP) module, a transformer-based conditional variational autoencoder (CVAE) is introduced. Finally, conditioned on the fused feature, actions are generated by the diffusion head.

pensate for sparsity in challenging tasks.

- We design a lightweight Object Position Prediction (OPP) module to predict object positions. By introducing rich trajectory priors to facilitate the training of the diffusion policy, this module enhances fine-grained manipulation capabilities with negligible computational overhead.

II. RELATED WORKS

A. Imitation Learning in Robotic Manipulation

Imitation learning aims to acquire the robotic skills from expert demonstrations. Behavioral cloning (BC) [23] is a classic imitation learning approach to directly mapping observations to actions, and formulates the policy learning as a supervised regression problem. Due to the compounding errors [24], BC cannot efficiently learn long-horizon tasks. To cope with this issue, new imitation learning methods such as implicit BC [25] and Transformer-based architecture [1] have been developed, in which the attention mechanism and action chunking are leveraged to capture action trajectories. Recently, the diffusion-based imitation learning [2–5] has gained great research attention as a new policy paradigm. Inspired by the success of achieving image generation and semantic scene synthesis tasks [26], diffusion policies iteratively denoise random noise into action sequence. Later work [27] supports fast sampling of diffusion policies for goal-specific action generation. However, effective observation encoder needs to be designed for precise manipulation when deploying the diffusion policies.

B. Visual Representation in Robot Learning

Visual representations are critical for robotic systems. Early research primarily uses large-scale datasets (e.g., ImageNet) to pre-train 2D CNNs [28] or vision transformers [29]. Recent works [30–32] study visual pre-training via video or language supervision to improve 2D policy generalization. Meanwhile, 3D representation learning enables spatial geometry understanding. Although voxels and NeRF facilitate geometric reasoning [6, 33], they incur high computational costs. Conversely, SE(3)-equivariant point cloud methods [3, 4, 34, 35] accelerate inference by directly processing sparse points.

To leverage both geometry and semantics, multimodal fusion is widely adopted [9, 13, 36, 37], primarily through naïve concatenation [10] or flow representations [38]. While conditioning mechanisms operate well in image generation [32, 39], integrating semantics and geometric information must be delicately tailored for efficient robotic perception.

C. Motion Guidance and Trajectory Prediction

The appropriate object-centric representation is essential in the robotic manipulation domain [40]. Compared with the end-to-end approaches that map pixels directly to actions, incorporating object state representations significantly enhances policy robustness [19, 41]. Early works [42, 43] have considered object 6D pose estimation for robotic manipulation. To reduce reliance on precise geometric models, the keypoint based methods such as [40, 44] localize semantic points of interest to simplify object state representation; the scene flow based methods such as [38] estimate the object surface normals to guide motion planning. Long-term point tracking technologies have been developed in [20, 45, 46], where complex motion of objects are captured through tracking sparse keypoints. These works highlight that compared to pursuing nearly complete pose estimation, learning low-dimensional trajectories or keypoint representations not only helps reduce computational burden, but also delivers adequate state information to guide manipulation policies.

III. METHOD

The proposed MOP is an object position guided multimodal policy as shown in Fig. 2. Differing from the existing multimodal imitation learning methods, the geometry-guided MOP is capable of avoiding the modality competition in an adaptive multimodal fusion manner. MOP consists of three components: i) Geometry-guided multimodal perception (G2MP) decouples 2D semantics and 3D geometric structure, to provide robust state representations; ii) Object position prediction (OPP) module estimates the object positions and thus provides the geometric priors; iii) Diffusion-based policy generates the precise robotic actions.

A. Geometry-Guided Multimodal Perception (G2MP)

In the unstructured environments, robotic perception needs to comprehensively understand both the object geometry and visual semantics. Therefore, we first adopt unimodal encoders for complementary strengths.

1) *Spatial Softmax based Position Encoder*: The encoder utilizes ResNet-18 based backbone. Different from the global average pooling that encodes the visual semantics into a 1D feature [13], spatial softmax pooling is adopted to maintain pixel position information. Assuming that each pixel corresponds to a 2D coordinate μ , and the image feature map extracted by the ResNet-18 is $\mathbf{f} \in \mathbb{R}^{C \times H \times W}$ with C feature channels and the spatial dimensions H, W , the keypoint coordinate μ_c for each feature channel c is as follows:

$$\mu_c = \sum_{h=1}^H \sum_{w=1}^W \mu(h, w) e^{f_{c,h,w}/\tau} / \left(\sum_{i=1}^H \sum_{j=1}^W e^{f_{c,i,j}/\tau} \right), \quad (1)$$

where $f_{c,h,w}$ is the activation for each feature channel c at $\mu(h, w)$, τ is the temperature. Thus we can transform the image feature into C keypoint coordinates denoted by $\mathbf{F}_v \in \mathbb{R}^{C \times 2}$.

2) *Feature Extraction from Point Clouds*: Given the observed point clouds $\mathbf{P} \in \mathbb{R}^{N \times 3}$, we use a lightweight DP3 encoder [4], namely the geometric feature $\mathbf{F}_g$ is obtained after a three-layer MLP and max-pooling operations.

3) *Gated Geometry-Query Fusion*: To address the modality competition issue for multimodal fusion, we propose a geometry-guided and geometry-query gated fusion mechanism. Specifically, the gated fusion feature $\mathbf{Z}_{\text{en}}$ is obtained in a residual manner:

$$\mathbf{Z}_{\text{en}} = \mathbf{F}_g + \alpha \tilde{\mathbf{F}}, \quad (2)$$

where the learnable gate α ensures that geometric feature is dominant, and $\tilde{\mathbf{F}}$ is expressed as:

$$\tilde{\mathbf{F}} = \text{Attention}(\mathbf{Q} = \mathbf{W}_Q \mathbf{F}_g, \mathbf{K} = \mathbf{W}_K \mathbf{F}_v, \mathbf{V} = \mathbf{W}_V \mathbf{F}_v), \quad (3)$$

in which $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ are the query, key and value, respectively, $\mathbf{W}_Q$, $\mathbf{W}_K$ and $\mathbf{W}_V$ are linear projections. During the training, Modality-Dropout is adopted to ensure that the model keeps geometry-dominated, i.e., with probability p , randomly zero out image features to force the network to learn robust geometric representations. When point cloud is adequate, the gate α automatically reduces reliance on semantic information. Conversely, the network actively extracts visual semantics to enhance feature expressiveness when the geometric information is inadequate. Finally, the gated fusion feature $\mathbf{Z}_{\text{en}}$ and the encoded robot state feature $\mathbf{Z}_{\text{robot}}$ are concatenated to form observation representations $\mathcal{O}$.

B. Object Position Prediction (OPP)

In the complex environments, the use of object motion information improves task execution. The work [19] employs a MoCap system for demonstration collection and presents MBA policy consisting of two cascaded diffusion processes for predicting the object poses and generating robot actions, which lead to high acquisition costs and inference delay in practice. As such, we introduce OPP, a lightweight CVAE based module for object position prediction without the assistance of MoCap system. During training, the latent variable z is generated by a Transformer-based encoder, whose input sequence consists of the observation representations $\mathcal{O}$ and the

ground-truth future object trajectory data. During inference, since the ground-truth future object trajectory is unavailable, we directly sample z from the learned prior distribution to capture the latent diversity of the predicted trajectories. In the Transformer-based decoder, the memory sequence $\mathbf{M}$ is formed by stacking $\mathcal{O}$ and z . The object query $\mathbf{Q}$ are derived from a set of independent learnable embeddings $\mathbf{E}_q$. These embeddings are randomly initialized and optimized end-to-end with the entire network via backpropagation. Through CVAE, OPP predicts future object position information of T_{obj} steps, namely $\mathcal{P} = \{\mathbf{p}_t\}_{t=1}^{T_{\text{obj}}}$ with $\mathbf{p}_t = (x_t, y_t, z_t)$, which is given by:

$$\mathcal{P} = \text{Attention}(\mathbf{Q} = \mathbf{W}_Q \mathbf{E}_q, \mathbf{K} = \mathbf{W}_K \mathbf{M}_K, \mathbf{V} = \mathbf{W}_V \mathbf{M}), \quad (4)$$

where $\mathbf{M}_K = \mathbf{M} + \mathbf{P}_{\text{mem}}$, $\mathbf{P}_{\text{mem}}$ represents the memory sequence positional encoding.

Incorporating $\mathcal{P}$ via residual gating yields the conditional inputs of the diffusion policy:

$$\mathcal{I}_{\text{policy}} = \mathcal{O} + \beta \cdot \text{MLP}(\mathcal{P}, \mathcal{O}), \quad (5)$$

where the perceptual feature gate β is learnable parameter. Compared to MBA with object pose information (a 9D-vector), our OPP module only predicts the object position, without requiring object pose labeled data.

C. Diffusion-based Policy Execution

In the execution module, the noise-free action sequences $\mathcal{A}_0$ are generated by leveraging a conditional diffusion model. Specifically, the observation features $\mathcal{I}_{\text{policy}}$ is injected into the noise prediction network ε_θ as the condition via Feature-wise Linear Modulation (FiLM), the denoising process at the ℓ -th step is performed as follows:

$$\mathcal{A}_{\ell-1} = \frac{1}{\sqrt{\alpha_\ell}} (\mathcal{A}_\ell - \frac{1 - \alpha_\ell}{\sqrt{1 - \bar{\alpha}_\ell}} \varepsilon_\theta(\mathcal{A}_\ell, \mathcal{I}_{\text{policy}}, \ell)) + \sigma_\ell \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (6)$$

where α_ℓ and σ_ℓ are the noise schedule [48], $\bar{\alpha}_\ell = \prod_{s=1}^\ell \alpha_s$.

D. Optimization Objective

The proposed MOP framework employs joint end-to-end training, to ensure that modules are coordinated. Therefore, the total loss function is calculated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{opp}} \mathcal{L}_{\text{opp}} + \lambda_{\text{reg}} \mathcal{L}_{\text{sparse}}, \quad (7)$$

where $\mathcal{L}_{\text{diff}}$ is the diffusion loss, $\mathcal{L}_{\text{opp}}$ is the OPP loss with the weight λ_{opp} and the sparsity loss regularizer $\mathcal{L}_{\text{sparse}}$ with the weight λ_{reg} .

1) *Diffusion Loss*: To train the noise prediction network ε_θ , we use MSE as the objective:

$$\mathcal{L}_{\text{diff}} = \mathbb{E} \left[\left\| \varepsilon^\ell - \varepsilon_\theta(\sqrt{\bar{\alpha}_\ell} \mathcal{A}_0 + \sqrt{1 - \bar{\alpha}_\ell} \varepsilon^\ell, \mathcal{I}_{\text{policy}}, \ell) \right\|_2^2 \right]. \quad (8)$$

2) *OPP Loss*: The CVAE-based OPP loss includes the trajectory reconstruction term and KL divergence term for regularizing the latent space:

$$\mathcal{L}_{\text{opp}} = \sum_{t=1}^{T_{\text{obj}}} \left\| \mathbf{p}_t - \mathbf{p}_t^{\text{GT}} \right\|_1 + \beta_{\text{KL}} D_{\text{KL}} \left(q(z | \mathcal{P}^{\text{GT}}, \mathcal{O}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}) \right), \quad (9)$$

where $\mathcal{P}^{\text{GT}} = \{\mathbf{p}_t^{\text{GT}}\}$ are the ground-truth object position information, β_{KL} is the weight and $q(z | \mathcal{P}^{\text{GT}}, \mathcal{O})$ is the probabilistic encoder.

TABLE II: Comparing MOP (ours) with the baselines across 56 tasks in 3 simulation environments. We run 3 seeds and report the average success rate (%) and standard deviation of the results.

Method	Adroit(3 tasks)	DexArt(4 tasks)	Meta-World Easy(28 tasks)	Meta-World Medium(11 tasks)	Meta-World Hard(5 tasks)	Meta-World Very Hard(5 tasks)	Average
DP [2]	25.8±4.0	40.3±2.4	81.2±2.5	52.9±4.0	47.9±5.0	54.3±4.2	64.4±3.3
DP3 [4]	74.9±1.4	51.1±3.3	84.8±3.0	63.9±4.4	53.2±6.0	65.5±5.5	73.2±3.7
Mamba Policy [47]	76.2±3.0	51.4±3.5	84.3±2.6	62.2±4.2	58.1±3.7	63.9±3.8	73.1±3.2
MBA [19]	80.8±2.4	54.4±4.2	85.3±1.7	65.2±4.7	61.3±5.2	67.9±4.4	75.2±3.1
MOP (ours)	81.3±2.3	52.7±3.5	85.6±1.5	78.6±2.6	73.0±3.6	72.1±2.6	79.3±2.2

3) *Sparsity Loss*: Since system robustness can be further enhanced by introducing a dual residual learning mechanism, we adopt the sparsity loss to regularize the geometric feature gate α and the perceptual feature gate β :

$$\mathcal{L}_{\text{sparse}} = \mathbb{E} \left[|\alpha|^2 + |\beta|^2 \right]. \quad (10)$$

IV. SIMULATION AND REAL-WORLD EXPERIMENTS

A. Simulation Experiment Setup

Simulation Benchmarks. We consider the total 56 manipulation tasks in 3 simulation environments for performance evaluation:

- Adroit [21]: A 28-DoF dexterous arm-hand manipulator solves the tasks in the MuJoCo physics simulator [49]. We select 3 challenging tasks including door opening, hammer and repositioning a pen.
- DexArt [7]: A 22-DoF dexterous arm-hand manipulator solves the tasks in the SAPIEN physical simulator [50]. Following [7], we consider 4 tasks including Faucet, Bucket, Laptop and Toilet. These tasks focus on the interactions with articulated objects.
- Meta-World [22]: A 7-DoF Sawyer arm with the gripper solves the tasks in the MuJoCo physics simulator. We consider 49 tasks, to evaluate broad task generalization.

Evaluation Metric. We run 3 seeds for each experiment with seed number 0, 42, 66, and evaluate 20 episodes to calculate the success rate at each seed. The average success rate of the top five checkpoints is reported. The settings for the 3 simulation benchmarks are detailed as follows:

- Adroit: 10 expert demonstrations per task; each model is trained for 3000 epochs and evaluated every 200 epochs.
- DexArt: 50 expert demonstrations per task; each model is trained for 3000 epochs and evaluated every 200 epochs.
- Meta-World: 10 expert demonstrations per task; each model is trained for 1000 epochs and evaluated every 100 epochs.

Baselines. To comprehensively exhibit the performance behavior of the proposed MOP method, we compare it with four strong diffusion-based imitation learning policies. These baselines cover different visual observations, model architectures and inference strategies:

- DP [2]: An image-based imitation learning method utilizes the CNN and transformer as the backbones.
- DP3 [4]: A point cloud based imitation learning method utilizes a light-weight Pointnet to obtain the 3D representations.

- Mamba Policy [47]: A state space model (SSM) based 3D policy introduces Mamba architecture as the denoising network backbone.
- MBA [19]: An object-centric imitation learning method adopts two cascaded diffusion processes to predict the object motion and generate object motion guided actions with the help of a MoCap system.

Simulation settings. We use a single NVIDIA RTX 4090 GPU to train all the models. The detailed hyperparameters for all the baselines and our MOP method are summarized in Table I. Since only the baseline MBA and our MOP require object motion/position data as training labels, the object motion/position prediction horizon is set to the same value to ensure a fair comparison between these two policies. Specifically, for simulation experiments, ground-truth object poses are directly obtained via the MuJoCo [49] and SAPIEN [50] APIs to provide training supervision.

TABLE I: Hyperparameters for all the baselines and our MOP method.

Hyperparameters	Value	Method
Observation horizon	2	All
Action prediction horizon	16	All
Execution horizon	8	All
Motion/position prediction horizon	16	MBA /MOP (ours)
Visual encoder output dimension	128	All 3D policies
Diffusion hidden dimensions	(512,1024,2048)	All diffusion policies
Diffusion steps (training/inference)	100/10	All
Batch size	128	All
Learning rate	1×10^{-4}	All
$\lambda_{\text{opp}}, \lambda_{\text{reg}}$	1, 0.01	MOP (ours)

B. Simulation Results

Table II shows clear differences that our MOP achieves higher average success rate (%) with lower standard deviation across most evaluated tasks in 3 environments than the baselines, which confirms the strong robustness and generalization of our approach. In particular, MOP improves the average success rates by at least 10% in the Meta-World tasks with the medium and hard levels of difficulty compare to MBA, and reaches the best performance in Adroit tasks.

To evaluate the learning efficiency of the proposed model, we sample 8 simulation tasks with different levels of difficulty from Meta-World benchmark. Fig. 3 confirms that MOP converges rapidly through leveraging the proposed G2MP and OPP modules, and outperforms the baselines across these tasks. In **sweep-into** and **hand-insert** tasks, unlike existing methods that suffer from low efficiency and limited accuracy, MOP significantly improves accuracy due to its capability of representations learning and efficient guidance for action generation. In **bin-picking** task with small object, point cloud downsampling in existing 3D policies [4, 19, 47] leads to the geometric information loss and thus reduces accuracy; our

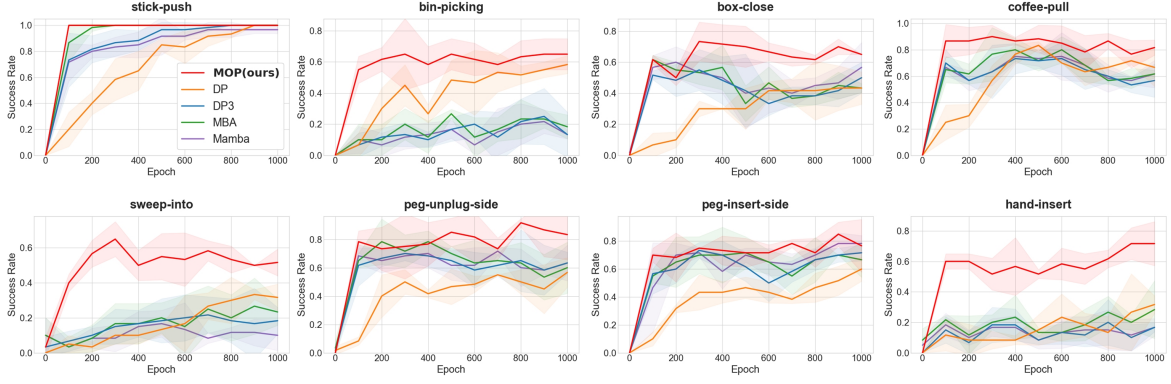


Fig. 3: **Learning curves of MOP and the baselines on Meta-World benchmark.** MOP achieves faster convergence and higher success rates than the baselines.

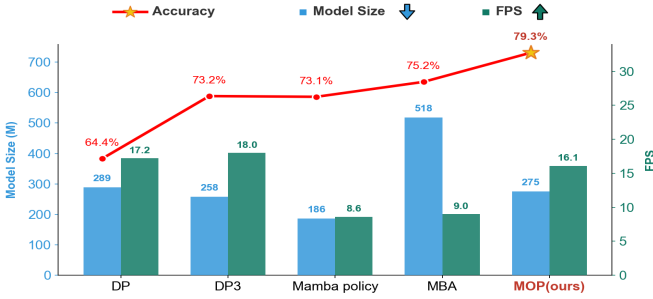


Fig. 4: **Effects of different models on the accuracy and inference speed (FPS) via a single NVIDIA RTX 4090 GPU.** MOP (ours) achieves the highest accuracy with reasonable inference speed.

MOP fuses 2D semantics with 3D features and can efficiently capture key spatial information in the complex scene.

Since the deployment of robot learning requires lower inference latency, high computational efficiency is also preferred. Fig. 4 shows that MOP achieves the highest accuracy (79.3%) by using 275M parameters, significantly less than MBA (518M parameters), and offers 1.8x inference speed compared to MBA. Notably, MOP maintains comparable inference efficiency to DP3; the added ResNet-18 (about 11M parameters) requires only a single forward pass, incurring minimal latency. Furthermore, our single-pass OPP module effectively avoids the iteratively denoising overhead seen in MBA. Consequently, with only a marginal drop in speed, MOP improves the accuracy by 6.1% over DP3 (258M parameters) and 14.9% over DP (289M parameters). Although Mamba Policy has the lowest number of model parameters (186M), its inference is the slowest (only 8.6 FPS) and accuracy (73.1%) is also decreased.

C. Real-world Experiments Setup

We conduct the real-world experiments across 4 manipulation tasks.

Platform. The real robot system consists of a 6-DoF UR3 arm with a Robotiq 2F-85 gripper and Orbbec Femto Mega RGB-D camera to capture observations, as shown in Fig. 5a). To train the OPP module, the object position labels are collected with the help of Cotracker (Fig. 5b)) [20]. Training and inference are performed by using a single NVIDIA RTX 4090 GPU and a single RTX 4060 GPU, respectively.

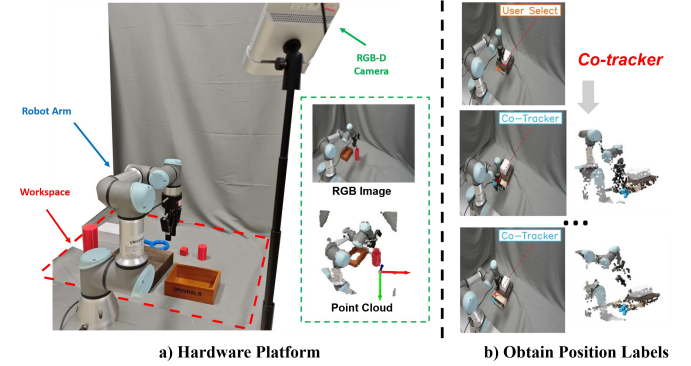


Fig. 5: **Real-world robot setup.** 6-DoF UR3 arm with a Robotiq 2F-85 gripper and Orbbec Femto Mega RGB-D camera to capture observations; Cotracker [20] is employed to obtain object position labels.

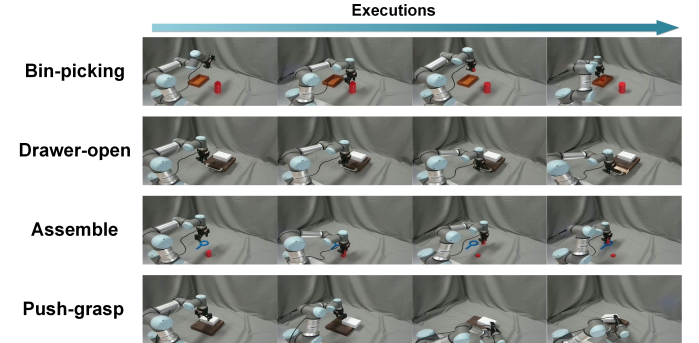


Fig. 6: Examples of execution steps across real-world tasks.

Baselines and Tasks. We compare MOP with 3D policies including DP3 [4] and Mamba policy [47]². In Fig. 6, we describe 4 real-world tasks:

- **Bin-picking** (Small object): Pick a small cubic object and place it into a bin. In this task, point cloud downsampling often leads to the loss of small object representation.
- **Drawer-open** (Articulated object): Open a drawer under random drawer positions. This task involves interactions with articulated object. Collision between the gripper and drawer needs to be reduced.

²As shown in Tables II and Fig. 4, our MOP outperforms MBA [19] in the simulation environments. Since MBA requires high-dimensional object motion data supported by a MoCap system for training, its real-world performance is not evaluated in our low-cost platform.

TABLE III: Real-world performance comparisons between **MOP** and the baselines across 4 manipulation tasks.

Method	Bin-picking		Drawer-open		Assemble		Push-grasp	
	Succ. (%) $\uparrow$	Collision $\downarrow$	Succ. (%) $\uparrow$	Grasp	Succ. (%) $\uparrow$	Succ. (%) $\uparrow$	Stage I Succ. (%) $\uparrow$	Succ. (%) $\uparrow$
DP3 [4]	20	11	30	70	30		40	30
Mamba Policy [47]	15	11	35	75	25		40	35
MOP (ours)	40	9	55	75	50		65	60

- **Assemble** (High-precision task): Grasp a cylindrical object with a diameter of 40mm and move it to the working space, and insert it into a circular ring with an inner diameter of 60mm.
- **Push-grasp** (multi-stage task): It consists of two stages: i) In Stage I, push a cuboid object to the edge until it overhangs; and ii) In the final stage, grasp it. This task inherently demands a narrow error tolerance: insufficient pushing results in a failed grasp, whereas excessive pushing drives the object off the platform.

We collect 50 expert demonstrations via teleoperation, and conduct 20 independent trials for each task.

D. Real-world Results

We evaluate MOP against DP3 and Mamba Policy on 4 complex tasks. Table III shows that MOP significantly improves the success rate across all the tasks, compared to the baselines. In **Bin-picking** task, MOP achieves a success rate twice that of DP3; In **Drawer-open** task, MOP improves the success rate by 20% over Mamba Policy. In **Assemble** task, our MOP enables highly precise manipulation, hence it improve the success rate by 20% over DP3. In addition, leveraging MOP increases the **Push-grasp** accuracy by at least 25% over Mamba Policy.

To further evaluate spatial generalization differences between MOP and the baselines including DP3 and Mamba Policy, we consider the experimental setup shown in Fig. 7, where 50 expert demonstrations are collected in **Area A** (inside the circle) for training. **Area A** and **Area B** (outside the circle) are seen and entirely unseen areas during training, respectively, and 20 independent trials are conducted in each area during testing. Table IV shows that the proposed MOP outperforms the baselines in both seen and unseen areas. However, all the methods exhibit a significant performance degradation when confronted with unseen **Area B**, due to the inherent generalization bottleneck of imitation learning. Compared to the baselines that cannot achieve any grasps, MOP successfully completes grasping at three locations of the unseen **Area B**, thanks to its better perceptual feature extraction and object position guidance.

TABLE IV: Spatial generalization performance for different policies.

Area	DP3 [4]	Mamba Policy [47]	MOP (ours)
	Succ. (%) $\uparrow$	Succ. (%) $\uparrow$	Succ. (%) $\uparrow$
Area A	50	45	65
Area B	0	0	15

E. Ablation Study

To analyze the effects of key modules in MOP, we conduct ablation experiments on 4 tasks in Meta-World environment, namely **coffee-pull (c-p)**, **sweep-into (s-i)**, **peg-unplug-side (p-u-s)** and **hand-insert (h-i)**. All the experiments follow the same simulation setting: All the models are trained across 3

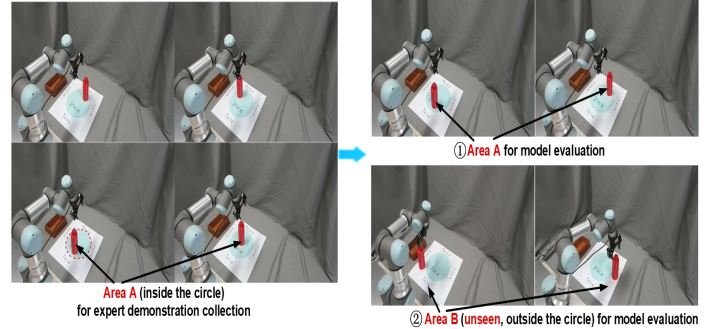


Fig. 7: **Experimental setup for spatial generalization evaluation.** Expert demonstrations are collected in **Area A** for training, and different policies are evaluated in **Area A** and entirely unseen **Area B**, respectively.

TABLE V: Ablation study on key modules in Meta-World environment.

Method	c-p	s-i	p-u-s	h-i	Average
MOP (ours)	91.0 $\pm$ 4.6	60.7 $\pm$ 1.5	90.3 $\pm$ 4.2	67.7 $\pm$ 5.1	77.4
MOP w/o OPP	87.0 $\pm$ 3.3	57.7 $\pm$ 2.4	82.7 $\pm$ 0.9	61.0 $\pm$ 6.2	72.1
MOP w/o RGB	75.0 $\pm$ 2.5	24.7 $\pm$ 2.9	77.7 $\pm$ 2.5	23.0 $\pm$ 2.9	50.1
MOP w/o gate α	80.7 $\pm$ 2.4	61.3 $\pm$ 1.3	59.3 $\pm$ 1.3	61.7 $\pm$ 3.3	65.8
MOP w/o gate β	86.0 $\pm$ 2.4	56.7 $\pm$ 3.1	83.0 $\pm$ 2.4	59.7 $\pm$ 5.0	71.4
DP3 [4]	73.0 $\pm$ 6.6	22.3 $\pm$ 8.4	72.0 $\pm$ 2.0	19.7 $\pm$ 6.1	46.8
MBA [19]	76.3 $\pm$ 5.1	23.7 $\pm$ 10.8	76.0 $\pm$ 4.4	26.3 $\pm$ 6.7	50.6

seeds; each model is trained for 1000 epochs, and 20 episodes are tested every 100 epochs. Besides the baseline methods DP3 and MBA, we create 4 MOP variants:

- **MOP w/o OPP**: In this variant, we exclude the OPP module while retaining the G2MP module. This design evaluates the policy’s performance without explicit object position and trajectory priors.
- **MOP w/o RGB**: In this variant, we remove the image encoder and the G2MP fusion module, reducing the policy to a unimodal point cloud baseline. This setting allows us to evaluate the differences between our lightweight OPP module and MBA with 9D object pose information.
- **MOP w/o gate α** : In this variant, we replace the α -adaptive gating module with straightforward concatenation, i.e., semantic and geometric features are simply concatenated, treating both modalities as equally informative.
- **MOP w/o gate β** : In this variant, we replace the β -adaptive gating module with straightforward concatenation. i.e., the output of OPP and output of G2MP are simply concatenated.

Our ablation study in Table V shows that overall, MOP can comprehensively exploit multimodal benefits and thus achieve the highest accuracy, compared to its variants and baselines. MOP w/o RGB performs similarly to MBA, which demonstrates that our lightweight OPP module can efficiently replace the expensive 9D object pose supervision approach and own fast inference speed. In addition, MOP improves the

accuracy by 11.6% and 6% over MOP w/o gate α and MOP w/o gate β , respectively, which confirms that the adaptive gated fusion mechanism introduced by MOP indeed addresses the modality competition.

F. Accuracy evaluation of object position prediction

The prior ablation studies have confirmed that introducing the proposed OPP module improves the accuracy. To further evaluate its trajectory prediction performance, we conduct an independent unit test. Specifically, we consider 4 tasks in Meta-World environment as illustrated in the subsection IV-E, and partition expert demonstration data into a training set and an entirely unseen test set at a ratio of 95% to 5%. For each task, model is trained for 1000 epochs and evaluated every 100 epochs, thus the best-performing model was ultimately selected for evaluation on the test set. In particular, since the baseline MBA predicts the 9D object pose, only the first three dimensions (i.e., the spatial coordinates) are extracted for fair comparison. TABLE VI shows that our proposed OPP module achieves higher accuracy than the MBA. This can be attributed to two factors: first, the proposed G2MP module provides OPP with rich perceptual representations; second, while MBA predicts 9D object pose information, OPP module only needs to forecast the object position, thereby obtaining precise object positions more efficiently.

TABLE VI: Root mean square error (RMSE) and mean absolute error (MAE) comparisons between **MOP** and the baseline MBA for trajectory prediction.

Method	c-p		s-i		p-u-s		h-i	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
MBA [19]	1.02	0.75	0.93	0.79	0.91	0.74	0.90	0.70
MOP (ours)	0.66	0.45	0.14	0.11	0.86	0.68	0.88	0.69

G. Analysis of Adaptive Gating Mechanism

To investigate the adaptive capabilities of the G2MP module across tasks with varying geometric features, we analyze two representative Meta-World tasks: **Window-close** (large objects, dense point clouds) and **Bin-picking** (small objects, sparse point clouds).

Fig. 8 reveals a task-dependent pattern in the gating parameter α , which controls the fusion weight of 2D image semantics. In the **Window-close** task, the target object exhibits large, continuous 3D surfaces. Since geometry-only methods (e.g., DP3) achieve near-perfect performance (100% success rate), the G2MP module recognizes the geometric sufficiency and keeps α extremely low (0.03). This allows MOP to maintain the high accuracy of point cloud-only methods in geometry-rich tasks.

Conversely, in the **Bin-picking** task, the small object's point cloud becomes extremely sparse after downsampling. This severely degrades the performance of geometry-only DP3 (22% success rate). Recognizing this geometric inadequacy, our G2MP module adaptively increases α to 0.2, injecting more visual semantics to compensate. This adaptive fusion boosts the success rate to 69%, significantly outperforming DP3. These results confirm that G2MP dynamically adjusts multimodal fusion weights based on geometric reliability, enabling robust manipulation in diverse environments.

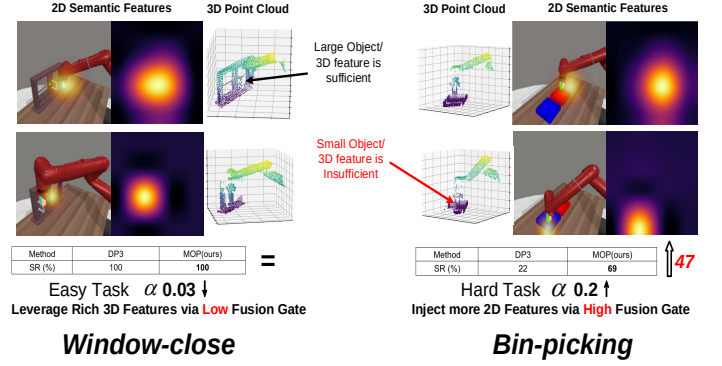


Fig. 8: **Analysis of adaptive gating mechanism.** The proposed G2MP module adaptively and efficiently fuses the 2D semantic features and geometric information, enabling significant success rate (SR) improvement in manipulation tasks involving small objects.

V. CONCLUSIONS

This work presented MOP, a novel multimodal object-aware policy designed for efficient and robust robotic manipulation. The proposed MOP incorporates a geometry-guided multimodal perception module, which leverages an adaptive gating mechanism to dynamically fuse 2D and 3D features. Moreover, we introduced a lightweight object position prediction module to inject rich trajectory priors into the policy without relying on external motion capture systems. Extensive experiments across 56 tasks in three simulation environments and 4 representative real-world tasks demonstrate that MOP outperforms strong baselines, exhibiting exceptional capability in handling challenging manipulation tasks.

In future work, the real-world performance boundaries of MOP will be further exploited across more diverse robots, dexterous hands, and tasks. Moreover, the line of this work will be extended to incorporate more control frameworks including combining imitation learning with reinforcement learning for enhancing generalization.

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